

Question 1

(a) Let $(x_n) \in V$. Then there exists $N \in \mathbb{N}$ such that $x_n = 0$ for all $n \geq N$. Then

$$\sup_{n \in \mathbb{N}} |T_n(x)| = \sup_{n \in \mathbb{N}} |n| \cdot \|e^n\| \cdot |x_n|$$

For $n \geq N$, $|x_n| = 0$, so it follows that

$$\sup_{n \in \mathbb{N}} |T_n(x)| = \sup_{\substack{n \in \mathbb{N} \\ n \leq N}} |n| \cdot \|e^n\| \cdot |x_n| \leq N \max_{\substack{n \in \mathbb{N} \\ n \leq N}} (\|e^n\|) \max_{\substack{n \in \mathbb{N} \\ n \leq N}} |x_n|$$

Since $\|e^n\|$ is finite, as $\|\cdot\|$ is a norm on V , these are maximums of finite sets, so they exist and are finite. Thus

$$\sup_{n \in \mathbb{N}} |T_n(x)| < \infty$$

(b) Suppose, for a contradiction, $(V, \|\cdot\|)$ is Banach. We have that each $T_n \in V'$ so each T_n is a linear continuous mapping. Thus, by the Banach-Steinhaus theorem,

$$\sup_{n \in \mathbb{N}} \|T_n\|_{V'} < \infty$$

For any $x = (x_n) \in V$, we have that

$$|T_n(x)| \leq \|x\|_V \cdot \|T_n\|_{V'}$$

So, for $x \neq 0$,

$$\sup_{n \in \mathbb{N}} \|T_n\|_{V'} \geq \sup_{n \in \mathbb{N}} \frac{|T_n(x)|}{\|x\|_V}$$

This is true for every $x \in V \setminus \{0\}$. But now, for each $N \in \mathbb{N}$, for $x = (e^N)$,

$$\frac{|T_N((e^N))|}{\|(e^N)\|} = \frac{|N| \cdot \|e^N\| \cdot |e_N^N|}{\|e^N\|} = N$$

But this implies

$$\sup_{n \in \mathbb{N}} \|T_n\|_{V'} = \infty$$

a contradiction. So $(V, \|\cdot\|)$ is not Banach.

Question 2

(a) First notice, for $\alpha, \beta \in \mathbb{K}$ and $L_1, L_2 \in V'$;

$$\Phi(\alpha L_1 + \beta L_2) = (\alpha L_1 + \beta L_2)|_M = \alpha L_1|_M + \beta L_2|_M = \alpha \Phi(L_1) + \beta \Phi(L_2)$$

So Φ is linear. On the other hand,

$$\|\Phi(L)\|_{M'} = \|L|_M\|_{M'} = \sup_{\substack{\|v\|_M=1 \\ v \in M}} \|L(v)\|_M = \sup_{\substack{\|v\|_M=1 \\ v \in M}} \|L(v)\|_V \leq \sup_{\substack{\|v\|_V=1 \\ v \in V}} \|L(v)\|_V = \|L\|_{V'}$$

So Φ is continuous and $\|\Phi\|_{\mathcal{L}(V', M')} \leq 1$. Now let $\ell : M \rightarrow \mathbb{K}$ be a nonzero linear continuous map. Then, by the Hahn Banach Theorem, there is a linear continuous map $L : V \rightarrow \mathbb{K}$ such that

$$L(v) = \ell(v), \quad \forall v \in M, \quad \text{and} \quad \|L\|_{V'} = \|\ell\|_{M'}$$

It thus follows that $\Phi(L) = \ell$, so

$$\|\Phi\|_{\mathcal{L}(V', M')} \geq \frac{\|\Phi(L)\|_{M'}}{\|L\|_{V'}} = \frac{\|\ell\|_{M'}}{\|L\|_{V'}} = 1$$

Thus $\|\Phi\|_{\mathcal{L}(V', M')} = 1$

- (b) Suppose Φ is injective. Assume that $\overline{M} \neq V$. Then $\overline{M}$ is a proper closed subspace of V . Thus for any $v \in V \setminus \overline{M}$, there exists some $L_v \in V'$ such that

$$L_v(u) = 0, \quad \forall u \in \overline{M}, \quad L_v(v) > 0$$

But then $0_{V'} \in V'$ and

$$\Phi(0_{V'}) = 0_{M'}, \quad \Phi(L_v) = 0_{M'}, \quad L_v \neq 0_{V'}$$

which contradicts that Φ is injective. So $\overline{M} = V$ and M is dense in V .

On the other hand, suppose M is dense in V and that $\Phi(L_1) = \Phi(L_2)$. Let $v \in V$. Then there is a sequence (m_n) in M converging to v . Thus

$$L_1(v) = L_1\left(\lim_{n \rightarrow \infty} m_n\right) = \lim_{n \rightarrow \infty} L_1(m_n)$$

Using the fact that L_1 is continuous. Then $\Phi(L_1) = \Phi(L_2)$ means $L_1(m_n) = L_2(m_n)$ and so

$$\lim_{n \rightarrow \infty} L_1(m_n) = \lim_{n \rightarrow \infty} L_2(m_n) = L_2\left(\lim_{n \rightarrow \infty} m_n\right) = L_2(v)$$

Thus $L_1 = L_2$ and Φ is injective.

- (c) Let $\ell \in M'$. By the Hahn Banach Theorem, there exists $L \in V'$ such that $L(v) = \ell(v)$ for all $v \in M$. But then $L|_M = \ell$ and so $\Phi(L) = \ell$. Thus Φ is surjective. Now V' and M' are both Banach spaces over the same field (dual spaces are always Banach), so, since we have shown Φ is surjective, by the Open Mapping Theorem, Φ is open.